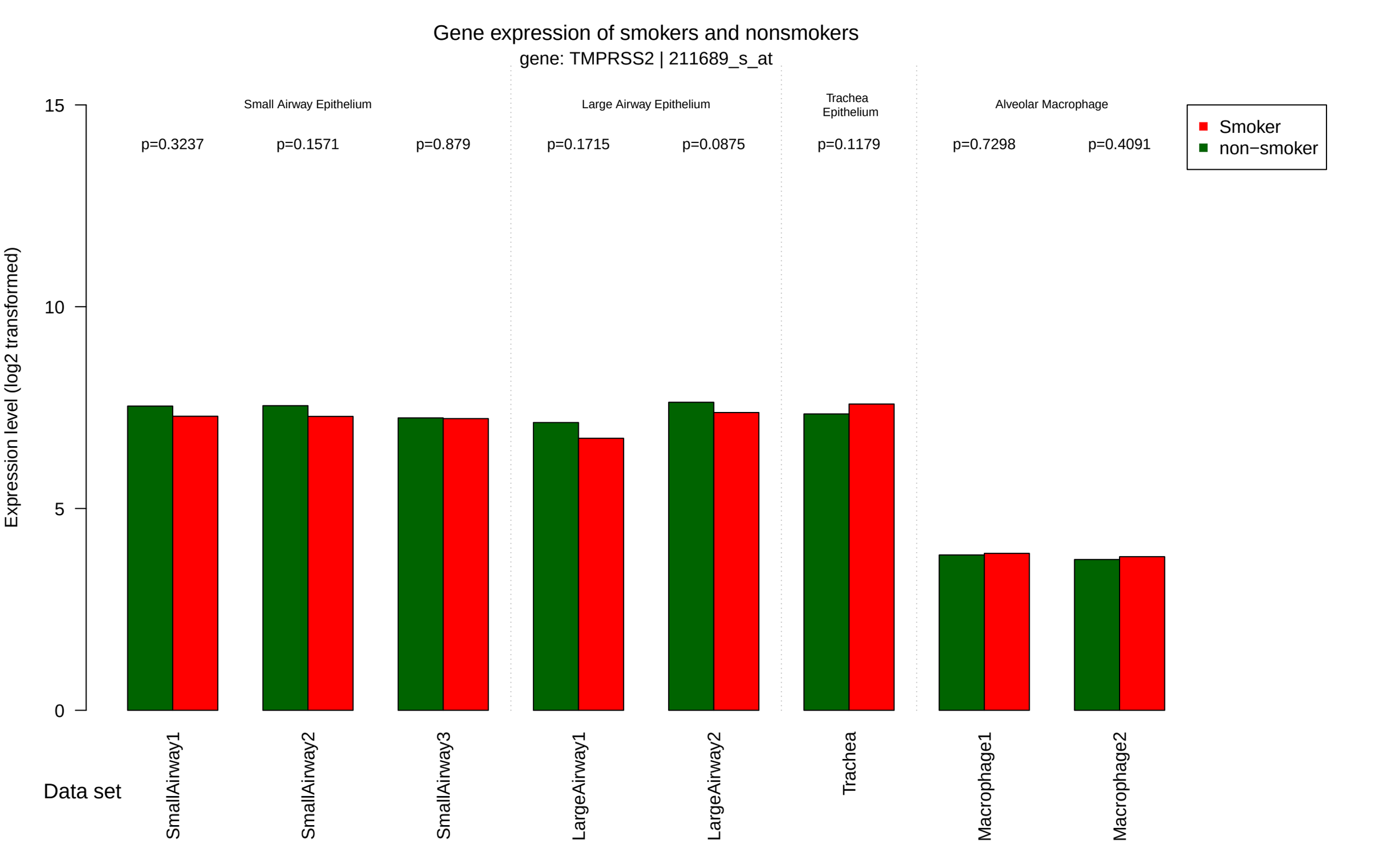
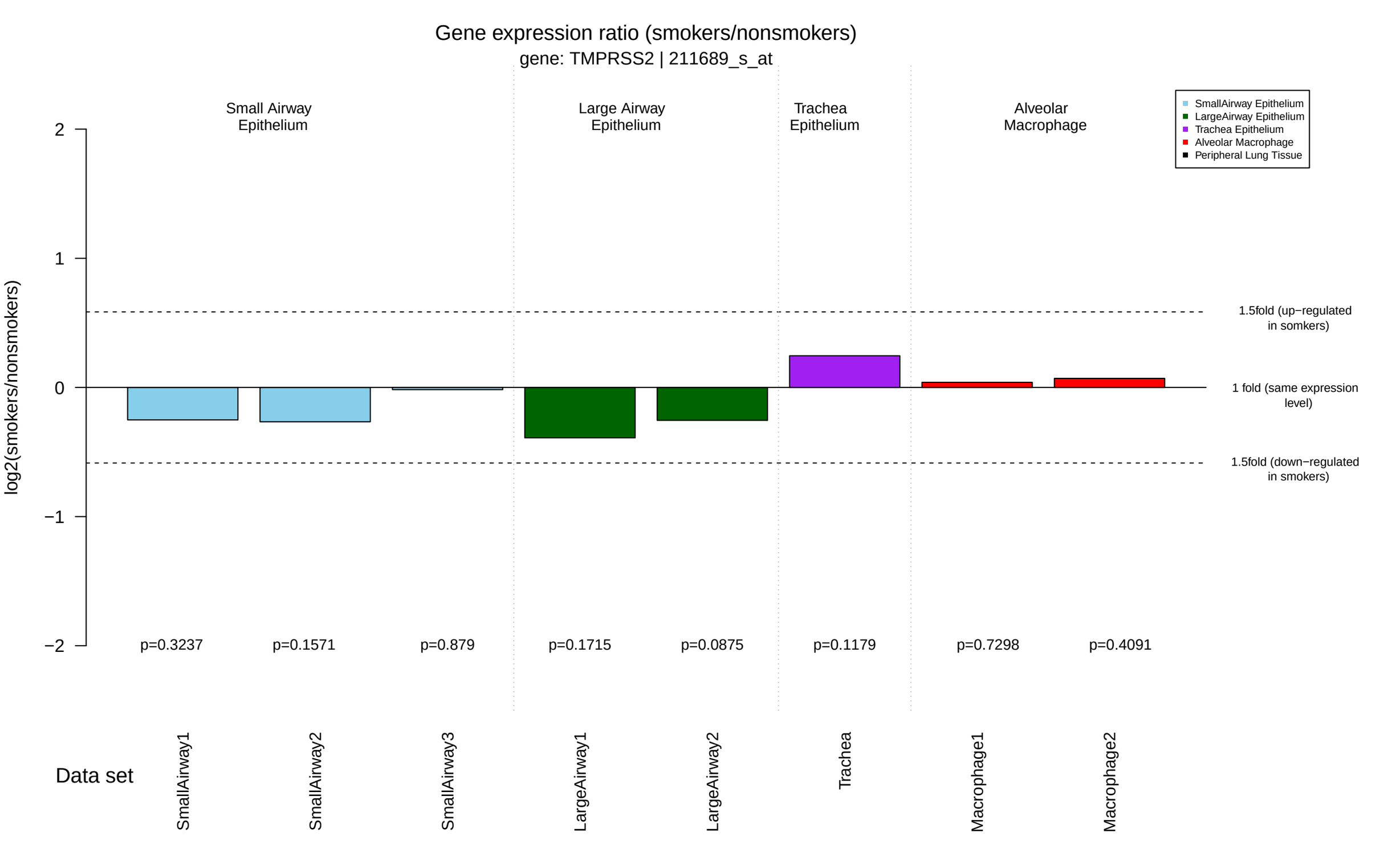
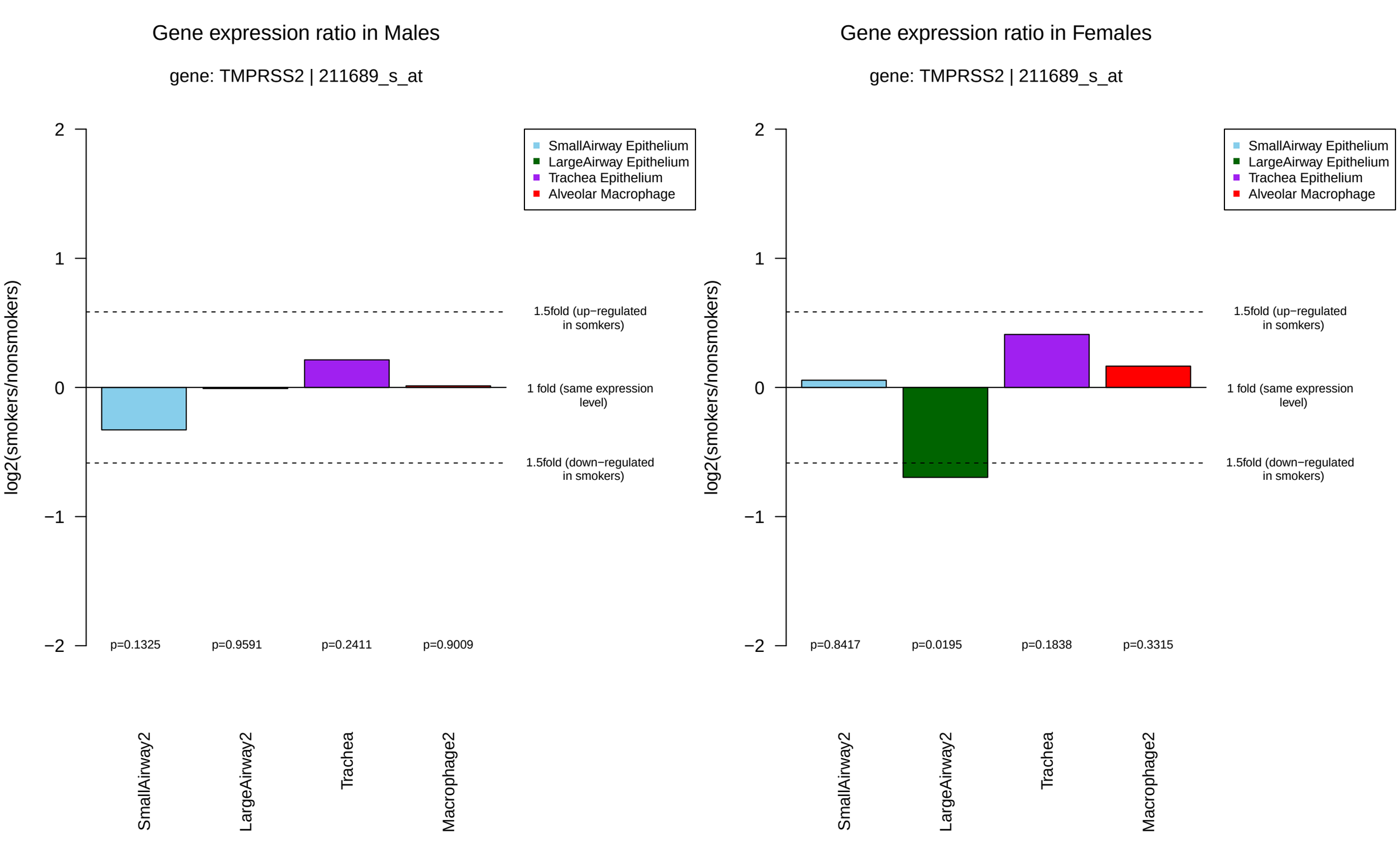
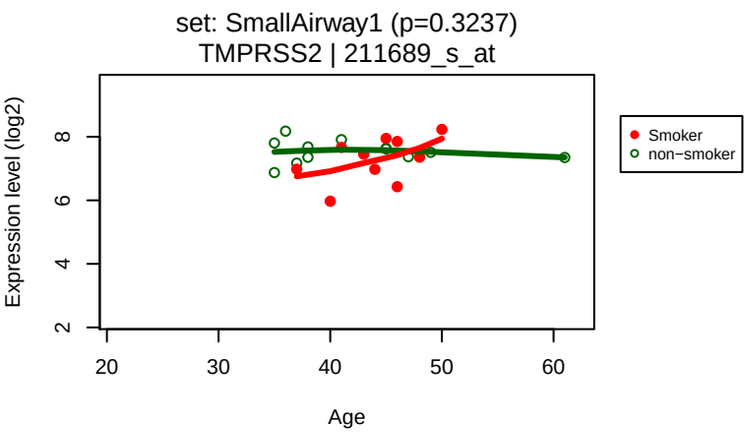
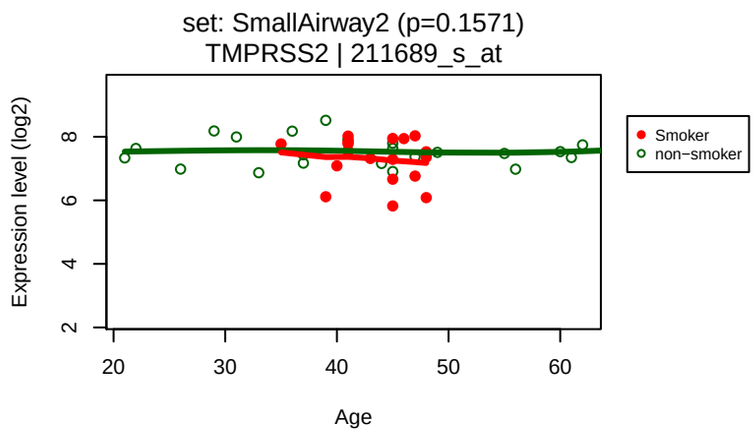
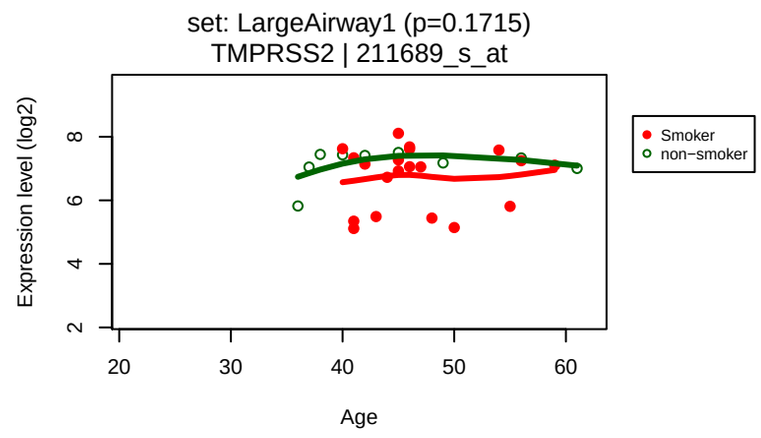

Gene expression of smokers and nonsmokers
gene: TMPRSS2 | 211689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3237
p=0.1571
p=0.879
p=0.1715
p=0.0875
p=0.1179
p=0.7298
p=0.4091
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

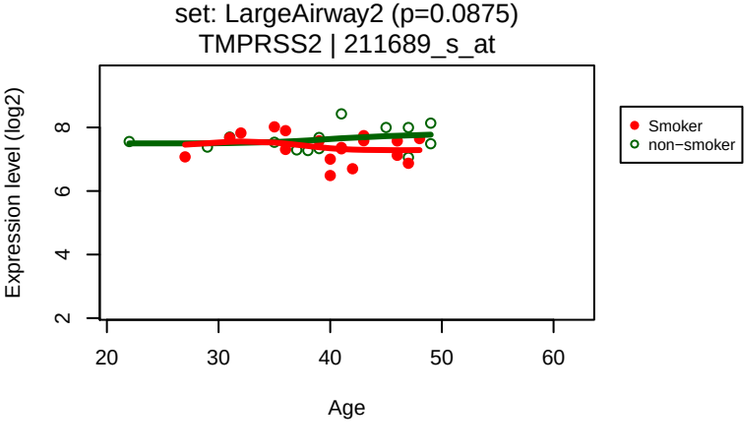

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS2 | 211689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3237
p=0.1571
p=0.879
p=0.1715
p=0.0875
p=0.1179
p=0.7298
p=0.4091
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

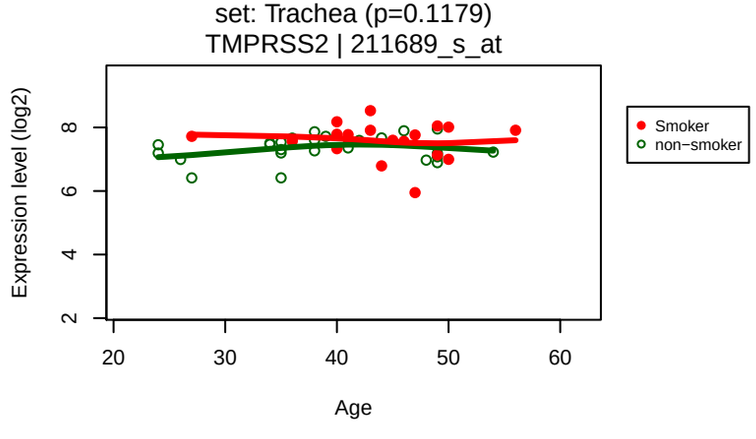

Gene expression ratio in Males
gene: TMPRSS2 | 211689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1325
p=0.9591
p=0.2411
p=0.9009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS2 | 211689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8417
p=0.0195
p=0.1838
p=0.3315
SmallAirway2
LargeAirway2
Trachea
Macrophage2

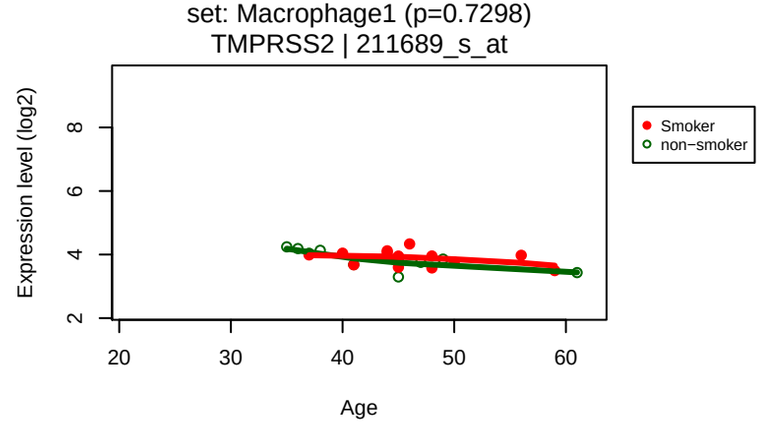

set: SmallAirway1 (p=0.3237)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker

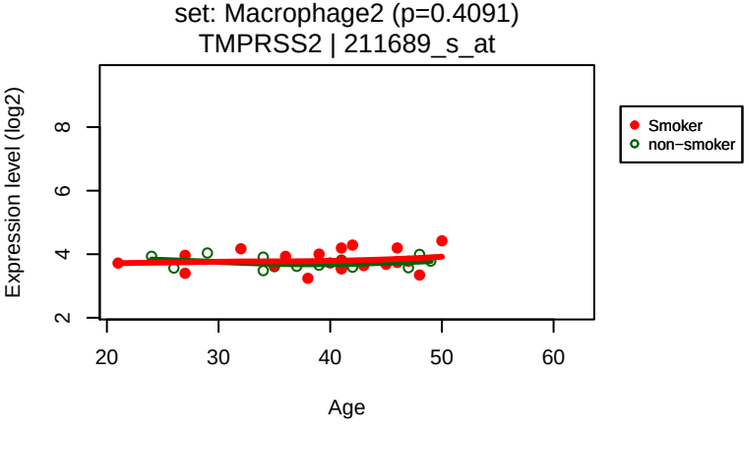

set: SmallAirway2 (p=0.1571)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker

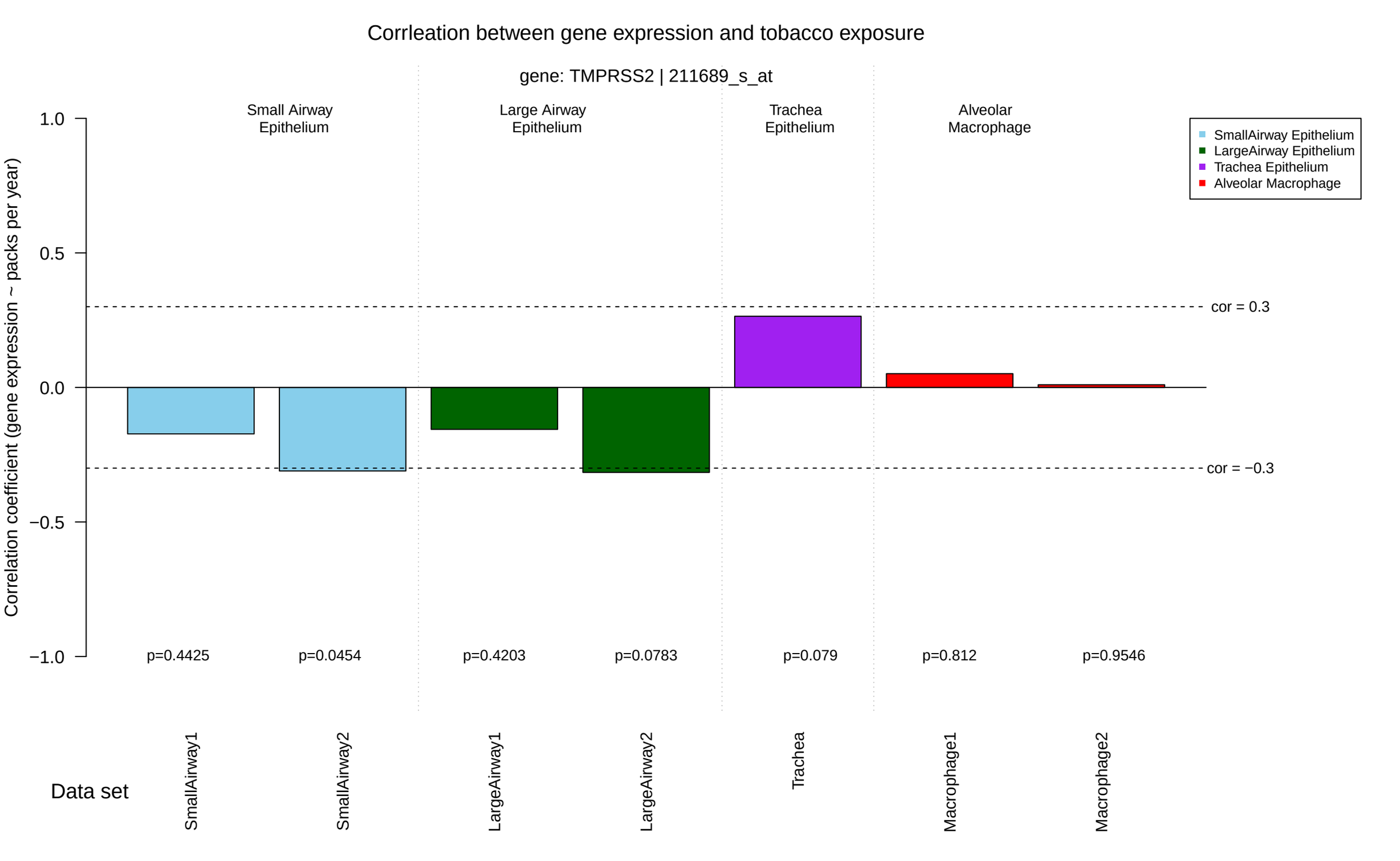

set: LargeAirway1 (p=0.1715)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0875)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1179)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7298)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4091)
TMPRSS2 | 211689_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS2 | 211689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4425
p=0.0454
p=0.4203
p=0.0783
p=0.079
p=0.812
p=0.9546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2